Chapter 6 Answer Key

Introduction to Statistics — MATH& 146

Section 6.1 Exercises

6.1.1. Hypotheses and test types:

(a) **Battery life.** $H_0 : \mu = 500$ hr vs. $H_a : \mu < 500$ hr. **Left-tailed.** (The consumer group suspects the true mean is below the claimed value.)

(b) **Reading scores.** $H_0 : \mu = 72$ vs. $H_a : \mu \neq 72$. **Two-tailed.** (The district only wants to know whether the mean has changed, with no prior direction specified.)

(c) **LDL reduction.** $H_0 : \mu = 30$ points vs. $H_a : \mu < 30$ points. **Left-tailed.** (The skeptic believes the true reduction is smaller than claimed.)

6.1.2. The null hypothesis must contain an equality because the p-value is computed by asking: "if H_0 is exactly true, how likely is our observed result?" That calculation requires a single specific value of the parameter to pin down the null sampling distribution. An inequality such as $\mu < 500$ does not specify a unique distribution; it encompasses infinitely many possible values of μ, so no single probability can be computed. The equality $\mu = \mu_0$ gives us the one distribution we need.

Section 6.2 Exercises

6.2.1. Test statistic $z = -1.88$. From the table: $P(Z \leq -1.88) = 1 - P(Z \leq 1.88) = 1 - 0.9699 = 0.0301$.

(a) **Left-tailed:** p-value $= P(Z \leq -1.88) = \mathbf{0.0301}$.

(b) **Right-tailed:** p-value $= P(Z \geq -1.88) = 1 - 0.0301 = \mathbf{0.9699}$.

(c) **Two-tailed:** p-value $= 2 \times 0.0301 = \mathbf{0.0602}$.

For each, sketch the standard normal curve, mark $z = -1.88$, and shade the appropriate tail(s). Note that the right-tailed p-value of 0.9699 is very large — $z = -1.88$ lies in the left tail, which is the opposite direction from a right-tailed alternative.

6.2.2. p-value $= 0.032$.

- **At $\alpha = 0.05$:** $0.032 < 0.05 \Rightarrow$ **reject** H_0. The data provide sufficient evidence at the 5% level.
- **At $\alpha = 0.01$:** $0.032 > 0.01 \Rightarrow$ **fail to reject** H_0. The same data do not provide sufficient evidence at the stricter 1% level.

The difference illustrates that the conclusion of a hypothesis test depends on the chosen threshold α. A more stringent significance level requires stronger evidence (a smaller p-value) before rejecting H_0.

6.2.3. The researcher has made two errors.

First, a p-value of 0.08 is *greater* than $\alpha = 0.05$, so the correct decision is to **fail to reject** H_0 — the data do not provide sufficient evidence against it. The researcher has the direction backwards.

Second, and more fundamentally, hypothesis testing **never proves H_0 true**. Failing to reject H_0 only means we lack sufficient evidence against it; it is entirely possible that H_0 is false but our sample was not large or informative enough to detect it. Absence of evidence is not evidence of absence. Corrected statement: "The data do not provide sufficient evidence at the 5% significance level to reject H_0. We fail to reject it, but we have not proven it true."

Section 6.3 Exercises

6.3.1. $H_0 : \mu = 16.0$ oz vs. $H_a : \mu > 16.0$ oz (right-tailed). $n = 50$, $\bar{x} = 16.08$, $\sigma = 0.2$.

(a) $\text{SE} = 0.2/\sqrt{50} \approx 0.02828$. Test statistic:

$$z = \frac{16.08 - 16.0}{0.02828} \approx \mathbf{2.83}.$$

(b) Right-tailed p-value:

$$p = P(Z \geq 2.83) = 1 - P(Z \leq 2.83) = 1 - 0.9977 = \mathbf{0.0023}.$$

(c) $p = 0.0023 < \alpha = 0.05$. **Reject H_0.** There is sufficient evidence at the 5% significance level to conclude that the machine is overfilling bottles (the true mean fill exceeds 16.0 oz).

6.3.2. Same data, $\alpha = 0.01$. Since $p = 0.0023 < 0.01$, we **still reject** H_0. The conclusion does not change: the evidence is strong enough to meet even the stricter 1% standard.

6.3.3. $H_0 : \mu = 100$, $z_{\text{obs}} = -1.75$.

From the table: $P(Z \leq -1.75) = 1 - P(Z \leq 1.75) = 1 - 0.9599 = 0.0401$.

(a) **Two-tailed p-value:** $p = 2 \times 0.0401 = \mathbf{0.0802}$.

(b) $p = 0.0802 > \alpha = 0.05$. **Fail to reject H_0.** Insufficient evidence to conclude $\mu \neq 100$.

(c) If $H_a : \mu < 100$ (left-tailed), then $p = 0.0401 < \alpha = 0.05$: **reject H_0.** Yes, the conclusion changes. The one-tailed test is more powerful in this direction because the entire significance threshold of 5% is concentrated in the left tail, rather than split between two tails. When we have a directional suspicion that is borne out by the data (the observed z is indeed negative), the one-tailed test picks it up more easily.

Section 6.4 Exercises

6.4.1. $H_0 : p = 0.60$ vs. $H_a : p < 0.60$. $n = 150$, $x = 81$. $\alpha = 0.05$.

Step 1. $\hat{p} = 81/150 = 0.54$. Conditions: $np_0 = 150(0.60) = 90 \geq 10$ ✓; $n(1 - p_0) = 150(0.40) = 60 \geq 10$ ✓.

Step 2.

$$\text{SE} = \sqrt{\frac{0.60 \times 0.40}{150}} = \sqrt{0.0016} = 0.0400. \qquad z = \frac{0.54 - 0.60}{0.0400} = \frac{-0.06}{0.040} = -1.50.$$

Step 3. Left-tailed: $p = P(Z \leq -1.50) = \mathbf{0.0668}$.

Step 4. $p = 0.0668 > \alpha = 0.05$. **Fail to reject H_0.** There is insufficient evidence at the 5% significance level to conclude that the true employment rate is less than 60%. The observed 54% could plausibly arise from sampling variation alone.

6.4.2. $H_0 : p = 0.50$ vs. $H_a : p \neq 0.50$ (fair coin test). $n = 100$, $x = 58$.

$\hat{p} = 0.58$. Conditions: $np_0 = 50 \geq 10$, $n(1 - p_0) = 50 \geq 10$ ✓.

$$\text{SE} = \sqrt{\frac{0.50 \times 0.50}{100}} = \sqrt{0.0025} = 0.05. \qquad z = \frac{0.58 - 0.50}{0.05} = 1.60.$$

Two-tailed: $p = 2(1 - P(Z \leq 1.60)) = 2(1 - 0.9452) = 2(0.0548) = 0.1096$.

$p = 0.1096 > 0.05$. **Fail to reject H_0.** There is insufficient evidence to conclude the coin is unfair. Getting 58 heads in 100 flips is not unusual enough to be convincing.

6.4.3. We need $|z| \geq 1.96$ to reject at $\alpha = 0.05$ (two-tailed). With SE $= 0.05$, the rejection boundary in the upper tail is:

$$\frac{\hat{p} - 0.50}{0.05} \geq 1.96 \implies \hat{p} \geq 0.598 \implies x \geq 59.8.$$

Since x must be a whole number, we need $x \geq 60$. Verification:

- $x = 59$: $z = (0.59 - 0.50)/0.05 = 1.80$, $p = 2(1 - 0.9641) = 0.0718 > 0.05$. *Do not reject.*
- $x = 60$: $z = (0.60 - 0.50)/0.05 = 2.00$, $p = 2(1 - 0.9772) = 0.0456 < 0.05$. *Reject.*

The minimum number of heads needed to reject H_0 (in the upper tail) is **60**.

Chapter 6 Review

1. Resting heart rate: $\mu_0 = 72$ bpm, $n = 45$, $\bar{x} = 75.8$, $\sigma = 12$.

(a) $H_0 : \mu = 72$ bpm vs. $H_a : \mu > 72$ bpm. **Right-tailed.** (The cardiologist suspects the rate is higher in high-stress workers.)

(b) SE $= 12/\sqrt{45} \approx 1.7889$.

$$z = \frac{75.8 - 72}{1.7889} \approx \mathbf{2.12}.$$

(c) Right-tailed p-value:

$$p = P(Z \geq 2.12) = 1 - P(Z \leq 2.12) = 1 - 0.9830 = \mathbf{0.0170}.$$

Sketch: standard normal curve with $z = 2.12$ marked; shade the right tail beyond 2.12.

(d)
- **At $\alpha = 0.05$:** $p = 0.0170 < 0.05$. **Reject H_0.** There is sufficient evidence at the 5% level that the mean resting heart rate for this occupation exceeds 72 bpm.
- **At $\alpha = 0.01$:** $p = 0.0170 > 0.01$. **Fail to reject H_0.** The evidence does not meet the stricter 1% threshold. The conclusion is sensitive to the choice of α.

2. Fast food calories: $\mu_0 = 540$, $n = 36$, $\bar{x} = 562$, $\sigma = 60$.
SE $= 60/\sqrt{36} = 60/6 = 10$.

(a) $H_0 : \mu = 540$ vs. $H_a : \mu \neq 540$ (two-tailed). $\alpha = 0.05$.

$$z = \frac{562 - 540}{10} = 2.20. \qquad p = 2(1-P(Z \leq 2.20)) = 2(1-0.9861) = 2(0.0139) = \mathbf{0.0278}.$$

$p = 0.0278 < 0.05$. **Reject H_0.** There is sufficient evidence that the true mean calorie count differs from the advertised 540 calories.

(b) **95% CI for μ:**

$$562 \pm 1.96 \times 10 = 562 \pm 19.6 \;\Rightarrow\; (542.4,\ 581.6) \text{ calories.}$$

The advertised value 540 is **not** inside $(542.4, 581.6)$.

(c) The test and CI are **consistent**: the hypothesis test rejected $\mu_0 = 540$ at $\alpha = 0.05$, and the 95% CI does not contain 540. This is exactly the duality of Section 6.5 in action — a two-tailed test at level α rejects μ_0 if and only if μ_0 lies outside the $(1 - \alpha)$ confidence interval.

3. Poll: $H_0 : p = 0.55$, $n = 200$, $x = 100$. $\hat{p} = 0.50$.
Conditions: $np_0 = 200(0.55) = 110 \geq 10$, $n(1 - p_0) = 200(0.45) = 90 \geq 10$ ✓.
SE $= \sqrt{0.55 \times 0.45/200} = \sqrt{0.0012375} \approx 0.03518$.
$z = (0.50 - 0.55)/0.03518 \approx -1.42$.
$P(Z \leq -1.42) = 1 - P(Z \leq 1.42) = 1 - 0.9222 = 0.0778$.

(a) $H_a : p < 0.55$ (left-tailed). p-value $= 0.0778 > 0.05$. **Fail to reject H_0.** Insufficient evidence to conclude support is below 55%.

(b) $H_a : p \neq 0.55$ (two-tailed). p-value $= 2(0.0778) = 0.1556 > 0.05$. **Fail to reject H_0.**

(c) The two-tailed p-value is exactly twice the one-tailed value because the standard normal curve is perfectly symmetric around zero. The one-tailed test shades only the left tail (area $= 0.0778$ beyond $z = -1.42$); the two-tailed test also shades the right tail beyond $|z| = 1.42$, which has the same area by symmetry, giving $2 \times 0.0778 = 0.1556$.

4. The journalist's sentence contains several statistical errors. A corrected version:

A p-value of 0.049 means that, *if the null hypothesis were true*, there would be about a 4.9% probability of observing a result as extreme as the one obtained. Because $0.049 < 0.05$, the researchers rejected the null hypothesis at the 5% significance level — this is not “95% confidence” (a term that belongs to confidence intervals, not p-values). More importantly, rejecting the null hypothesis does not “prove the hypothesis true”: it establishes that the data are inconsistent with the null at the chosen significance level, lending support to the alternative hypothesis; the research hypothesis may still be wrong, and the result could be a Type I error (a false rejection). Science does not prove; it provides evidence, and a single p-value just below 0.05 is a relatively weak form of it.